

CHAPTER 15

Vibrations

Key Features

After study of the subject matter presented in this chapter, the reader will acquire knowledge about :

- 15.1 Introduction
- 15.2 Concept and Type of Vibrations
- 15.3 Free Undamped Longitudinal Vibrations
- 15.4 Logarithmic Decrement
- 15.5 Free Vibration with Viscous Damping
- 15.6 Forced Damped Vibrations
- 15.7 Vibration Isolation
- 15.8 Summary

15.1 Introduction

Engineering system possessing mass and elasticity are capable of relative motion. Usually, vibrations are due to elastic forces. Vibration, in general, is a form of wasted energy and undesirable in many cases. This is particularly true in machinery; for it generates noise, breaks down parts. The failure of major structures such as bridges, buildings or airplane wings is an awesome possibility under resource. Vibration is the oscillation of structural or machine components rapidly and continuously under the excitation of forces which may be inherent in the component or applied externally. Sometimes, vibration may be beneficial. Various musical instruments are based on vibration of strings. Safety of the machine and supporting structure may be endangered due to vibration.

15.2 Concept and Type of Vibrations

Regular oscillatory or periodic motion which repeats itself after a definite interval is called vibration. Oscillation is the repetitive variation, typically in time, of some measure about a central value (often a point of equilibrium) or between two or more different states. Vibration refers to mechanical oscillation about an equilibrium point.

The main causes of setting up vibratory motion in a body are :

- (i) Unbalanced centrifugal forces
- (ii) Elastic nature of system.
- (iii) External excitations
- (iv) Winds
- (v) Dry friction between mating surfaces

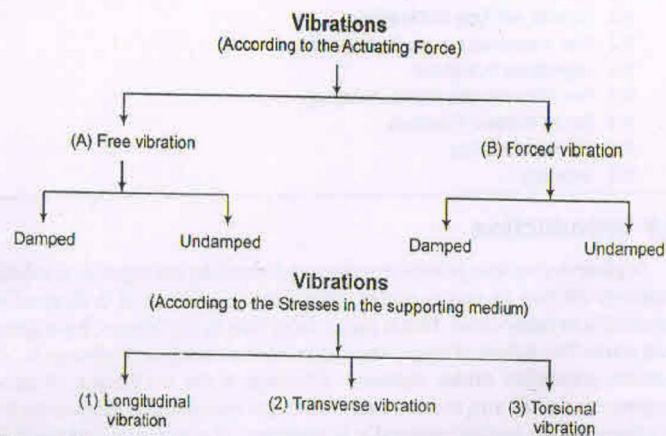
Advantages of vibration are :

- Vibrations are employed to simulate earthquakes for geological investigations.
- Vibration may be used for drilling of geo-technical wells.
- Vibrations are also employed in agriculture for harvesting by forced vibrations of fruit bearing trees.

Disadvantages of vibrations are :

- Unwanted vibrations produced unpleasant stresses in machine parts.
- Many structures, buildings and bridges falls because of vibration.
- The vibration causes rapid wear of machine parts such as bearings and gears.
- Excessive vibration is also dangerous for human beings.

15.2.1 Types of Vibration



According to the actuating force vibrations are :

(A) Free vibration : When after giving the initial displacement, external force is removed, and then body vibrates on its own.

(B) Forced vibration : Vibrations which are caused and maintained by a periodic disturbing force.

Difference between Free and Forced Vibrations

Free Vibration	Forced Vibration
1. Free vibration takes place without the application of external force.	1. Forced vibration takes with the application of external force
2. The frequency with which the system is vibrating is called natural frequency	2. The frequency with which the system is vibrating is called forced frequency
3. The natural frequency is a property of a system and it is constant for a particular system	3. There can be number of forced frequencies for a particular system

1. Longitudinal Vibrations : Consider a body of mass m as shown in Fig. 15.1 (a) carried on one end of a weightless spindle, the other end being fixed. If the mass moves up and down resulting in direct-tension and compression of spindle, it is said to execute Longitudinal vibrations.

In this case the different particles constituting the body move in parallel planes as shown in Fig. 15.1 (a).

2. Transverse Vibrations : If the body instead of moving vertically up and down vibrates in the exaggerates manner as shown in Fig. 15.1 (b) due to bending of the supporting medium, it is said to execute transverse vibrations.

In this case the particles of shaft or disc move approximately perpendicular to the axis of the shaft as shown in Fig. 15.1 (b).

3. Torsional Vibrations : When shaft gets twisted and untwisted alternatively due to vibratory motion of suspended body, it is said to be undergoing torsional vibration.

In this case shaft is subjected to torsional shear stresses and shaft particles move in a circle about the axis of shaft as shown in Fig. 15.1 (c).

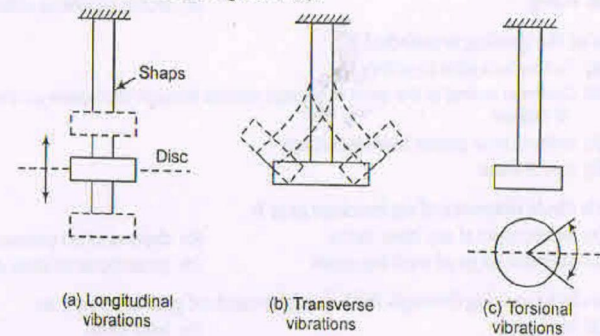


Figure 15.1

15.2.2 Important Definitions used in Vibration

- 1. Inertia :** The property of a body (either in rest or in motion) by which it continues to be in its present state unless acted upon by an external force.
- 2. Displacement (x) :** The change in position of an object in a particular direction by application of an external force.
- 3. Disturbance :** Any action which destroys the static equilibrium of a vibrating system may be called a disturbance to that system.
- 4. Restoring Force :** The displaced body does not stay in the new position because of the restoring force which is provided either by gravity or by elasticity or by both.
- 5. Undamped Vibration :** If vibrations take place in the absence of a damping force then the vibration is known as undamped vibration.
- 6. Damping :** It is resistance offered to the motion of the vibratory body.

7. **Damped Vibration** : If energy is lost or dissipated during oscillation, the vibration is known as damped vibration.
8. **Periodic Motion** : Motion which repeats itself at equal intervals of time.
9. **Inertia Force** : Inertia force is a resisting force to oppose the motion or change of motion in a straight line.

$$\text{The linear inertia force} = m\ddot{x}$$

where $\ddot{x}$ = linear acceleration.

The inertia force is assumed to be acting at the mass centre of the body in a direction opposite to the direction of acceleration.

10. **Inertia Torque** : Inertia torque is assumed to be acting at the body in the plane of motion in a direction opposite to the direction of angular acceleration of the body.

$$\text{Inertia torque} = I\ddot{\theta}$$

where $\ddot{\theta}$ = angular acceleration

15.2.3 Equivalent Stiffness of Spring Combinations

The stiffness members of vibrating system is represented by a coil spring. The multi-degree of freedom systems may be represented by more than one spring. It is necessary to find out equivalent stiffness of spring configuration for mathematical modelling.

The springs may be arranged in the following configurations,

1. Springs in series
2. Springs in parallel
3. Inclined springs

(i) **Springs in series** : Two springs of stiffness k_1 and k_2 are in series and supporting the same load, mg . The total static deflection of the equivalent spring must be equal to total static deflection of individual springs as shown in Fig. 15.2.

$$\Delta_{st} = \frac{mg}{k_e} = \frac{mg}{k_1} + \frac{mg}{k_2}$$

$$\frac{1}{k_e} = \frac{1}{k_1} + \frac{1}{k_2} = \frac{k_2 + k_1}{k_1 k_2}$$

$$k_e = \frac{k_1 k_2}{k_1 + k_2}$$

For n number of spring in series

$$\begin{aligned} \frac{1}{k_e} &= \frac{1}{k_1} + \frac{1}{k_2} + \frac{1}{k_3} + \dots + \frac{1}{k_n} \\ &= \frac{k_1 + k_2 + k_3 + \dots + k_n}{k_1 k_2 k_3 \dots k_n} \\ k_e &= \frac{k_1 k_2 k_3 \dots k_n}{k_1 + k_2 + k_3 + \dots + k_n} \end{aligned}$$

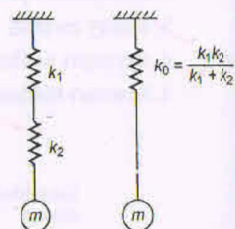


Figure 15.2 Springs in series

(ii) **Springs in parallel** : Two cases of single-degree of freedom system are shown in Fig. 15.3, where each spring is carrying a part of total load mg .

Total load supported by equivalent spring must be equal to the sum of the loads carried by individual springs.

$$mg = m_1 g + m_2 g$$

Total deflection of the system

$$\begin{aligned} \Delta_{st} &= \frac{mg}{k_e} = \frac{m_1 g}{k_1} = \frac{m_2 g}{k_2} \quad \text{Figure 15.3 Spring in parallel} \\ &= \frac{m_1 g + m_2 g}{k_1 + k_2} = \frac{g(m_1 + m_2)}{k_1 + k_2} = \frac{mg}{k_1 + k_2} \end{aligned}$$

$\therefore$

$$k_e = k_1 + k_2$$

For n number of springs in parallel,

$$k_e = k_1 + k_2 + k_3 + \dots + k_n$$

(iii) **Inclined springs** : If a mass m is connected to an inclined spring as shown a displacement of mass x in horizontal direction will result in spring deflection as $x \cos \phi$ in the horizontal direction as shown in Fig. 15.4.

The spring force in the direction of spring $= kx \cos \phi$. The component of the above spring force in the direction of displacement horizontal

$$= kx \cos \phi \times \cos \phi = kx \cos^2 \phi$$

The effective stiffness of spring, $k_e = k \cos^2 \phi$.

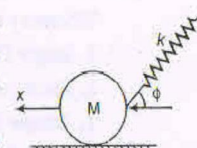


Figure 15.4 Inclined spring

15.2.4 Parts of Vibrating System

The elementary parts of vibrating system are mass, spring and damper (dashpot), which represent the body, the elasticity and the friction of the system respectively.

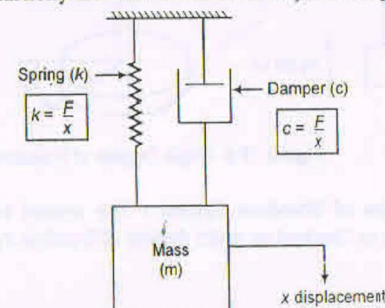


Figure 15.5

Expt: A vibration sys consists of a mass m & spring stiff-ness k . It has a natural freq of 12 Hz. The natural freq decreases by 2 Hz, when an extra mass of 2 kg attached to m . Find k & m .

Soln:

$$f_1 = \frac{1}{2\pi} \sqrt{\frac{k}{m}} = 12 \text{ Hz}$$

$$f_2 = \frac{1}{2\pi} \sqrt{\frac{k}{m+2}} = 12 - 2 = 10 \text{ Hz}$$

$$\text{as } \frac{k}{m} = (2\pi f_1)^2 = (2\pi \times 12)^2$$

$$\frac{k}{m+2} = (2\pi f_2)^2 = (2\pi \times 10)^2$$

$$\frac{m+2}{m} = \frac{(2\pi \times 12)^2}{(2\pi \times 10)^2} = 1.44$$

$$\text{or } 1.44m - m = 2$$

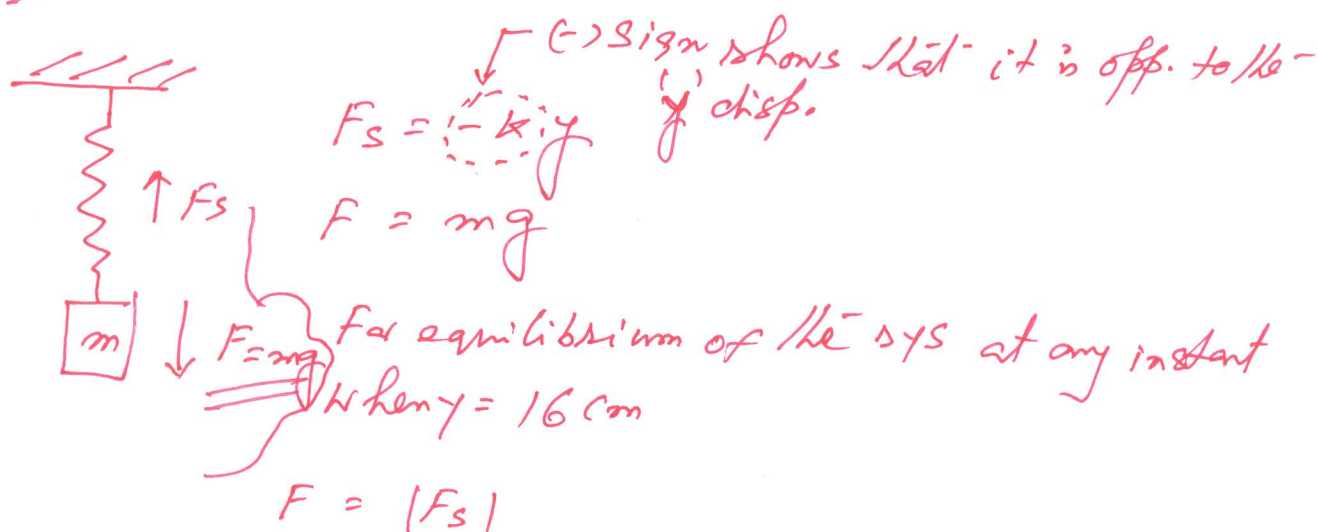
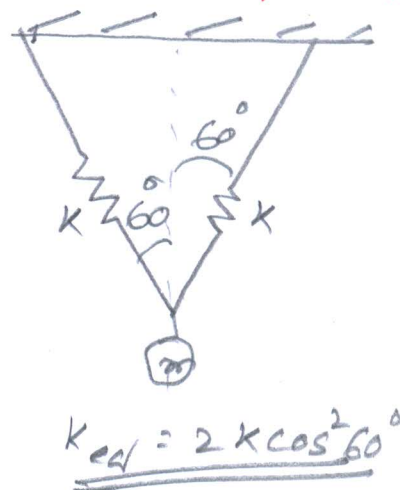
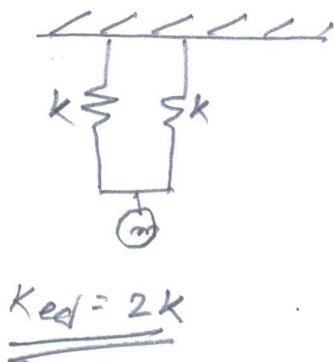
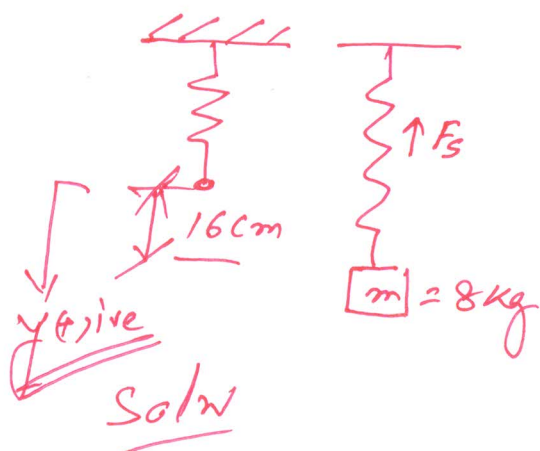
$$\therefore m = \frac{2}{0.44} = 4.54 \text{ kg}$$

$$\frac{k}{m} = (2\pi \times 12)^2$$

$$\therefore k = (2\pi \times 12)^2 \times 4.54$$

$$= 25783.25 \text{ N/m} = 25.783 \text{ N/mm}$$

Q. An 8 kg mass hangs from a spring & stretches by 16 cm beyond its natural length. Determine value of spring constant 'k' & the natural freq of the system.

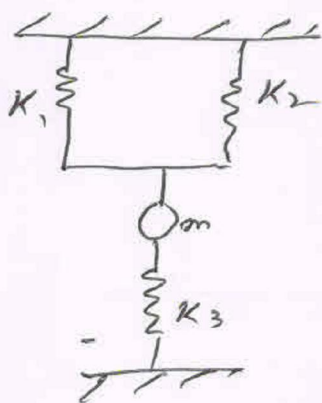


$$\text{or } mg = |-ky|$$

$$\text{or } \frac{mg}{y} = \left| \frac{-ky}{y} \right| \Rightarrow \frac{8 \times 9.81}{0.16} = k = 490.5 \text{ N/m}$$

$$f_n = \frac{1}{2\pi} \sqrt{\frac{k_{eq}}{m}} = \frac{1}{2\pi} \sqrt{\frac{490.5}{8}} = 1.25 \text{ Hz}$$

Exp: Find the natural freq of the system as shown



If $K_1 = K_2 = 1500 \text{ N/m}$, $K_3 = 2000 \text{ N/m}$
 $m = 5 \text{ kg}$

Soln: $K_{12} = K_1 + K_2$ (parallel springs)
 $= 2 \times 1500 = 3000 \text{ N/m}$

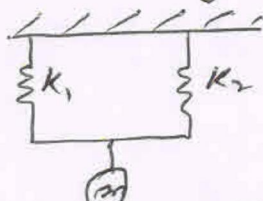
$\frac{1}{K_{123}} = \frac{1}{K_{12}} + \frac{1}{K_3}$ (Series)

$\frac{1}{K_{123}} = \frac{1}{3000} + \frac{1}{2000} \therefore K_{123} = \frac{3000 \times 2000}{3000 + 2000} = 1200 \text{ N/m}$

$\therefore f_n = \frac{1}{2\pi} \sqrt{\frac{K_e}{m}} = \frac{1}{2\pi} \sqrt{\frac{1200}{5}} = 2.68 \text{ Hz}$

Exp: A spring mass (k, m_1) has a natural frequency of f_1 . Another spring K_2 is connected in parallel to K_1 & the natural frequency increases by 30%. Find K_2 .

Soln:



$f_1 = \sqrt{\frac{K_1}{m_1}}$, $K_e = K_{12} = K_1 + K_2$ (parallel)

$f_2 = \sqrt{\frac{K_1 + K_2}{m_1}} = 1.3 f_1$

$\therefore \frac{f_1}{f_2} = \frac{1}{1.3} = \sqrt{\frac{K_1}{m_1} \cdot \frac{m_1}{K_1 + K_2}} \Rightarrow \frac{1}{1.3} = \sqrt{\frac{K_1}{m_1} \cdot \frac{m_1}{K_1 + K_2}}$
 $\text{or } \frac{1}{1.3} = \sqrt{\frac{K_1}{K_1 + K_2}}$

$\text{or } 1.69 K_1 = K_1 + K_2$

$\therefore K_2 = 0.69 K_1$

Exp: A circular cylinder of mass m & radius R is connected by a spring of stiffness k as shown. If it is free to roll on the rough surface which is horizontal without slipping. Find the natural freq of the system. What will happen to natural frequency if the sys is placed on incline as shown.

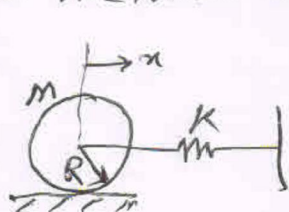


Fig-2

Soln: Let θ be the angular displacement for the linear distance x . Applying Newton's law for the cyl

$$m\ddot{x} = -kx + F \quad (F \text{ is frictional force})$$

but $I_0 \ddot{\theta} = -FR$ & $\left[\frac{1}{2}mR^2\right]\left[\frac{\ddot{x}}{R}\right] = -FR$

or $\theta = \frac{x}{R} \Rightarrow \ddot{\theta} = \frac{\ddot{x}}{R} \Rightarrow F = -\frac{1}{2}m\ddot{x}$

$$\therefore m\ddot{x} + kx + \frac{1}{2}m\ddot{x} = 0$$

$$\text{or } \ddot{x} + \left(\frac{2k}{3m}\right)x = 0$$

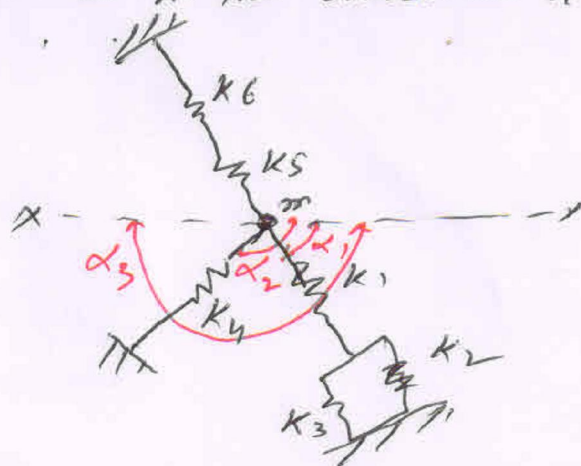
$$\therefore \omega_n = \sqrt{\frac{2k}{3m}} \text{ rad/sec}$$

$$f_n = \frac{1}{2\pi} \sqrt{\frac{2k}{3m}} \text{ cycles per sec } \text{Ans}$$

Fig-2

When the cyl is placed on incline no change in natural freq of the sys will occur.

Expt: Find the equivalent spring constant & fm for oscillation of mass m in the direction $x-x$ as shown.



Soln: $K_{23} = K_2 + K_3$ (parallel) $\left| \begin{array}{l} K_{123} = \frac{K_1 K_{23}}{K_1 + K_{23}} \\ \frac{1}{K_{123}} = \frac{1}{K_{23}} + \frac{1}{K_1} \text{ (Series)} \\ = \frac{K_1 (K_2 + K_3)}{K_1 + K_2 + K_3} \end{array} \right.$

Again $\frac{1}{K_{56}} = \frac{1}{K_5} + \frac{1}{K_6}$ (Series) $\therefore K_{56} = \frac{K_5 K_6}{K_5 + K_6}$

$\therefore$ Equivalent spring stiffness

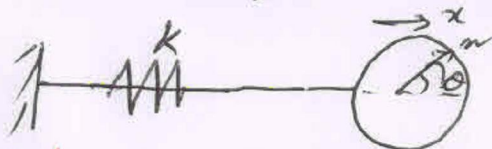
$$K_e = K_{123} \cos^2 \alpha_1 + K_4 \cos^2 \alpha_2 + K_{56} \cos^2 \alpha_3$$

$$= \frac{K_1 (K_2 + K_3)}{K_1 + K_2 + K_3} \cos^2 \alpha_1 + K_4 \cos^2 \alpha_2 + \frac{K_5 K_6}{K_5 + K_6} \cos^2 \alpha_3$$

The equation of motion of mass m is

$$m\ddot{x} + K_e x = 0 \quad \& \quad f_n = \frac{1}{2\pi} \sqrt{\frac{K_e}{m}}$$

Exp: A circular cyl of mass 4 kg & radius 15 cm is connected to a fixed support thru a spring of $k = 4000 \text{ N/m}$ as shown. The cyl is free to roll on a smooth horizontal surface without slip. Find f_n of the sys.



Soln: $x = R\theta$

Total energy of the sys = K.E. of translation motion of cyl + K.E. of rotary motion of cyl + P.E. of the spring due to strain.

$$= \frac{1}{2} m \dot{x}^2 + \frac{1}{2} I \dot{\theta}^2 + \frac{1}{2} k x^2 = \text{Constant}$$

$$= \frac{1}{2} m R^2 \dot{\theta}^2 + \frac{1}{2} \times \frac{1}{2} m R^2 \dot{\theta}^2 + \frac{1}{2} k R^2 \theta^2 = \text{Constant}$$

$$\therefore \frac{d}{dt} \left[\frac{1}{2} m R^2 \dot{\theta}^2 + \frac{1}{4} m R^2 \dot{\theta}^2 + \frac{1}{2} k R^2 \theta^2 \right] = 0$$

$$\therefore m R^2 \dot{\theta} \ddot{\theta} + \frac{1}{2} m R^2 \dot{\theta} \ddot{\theta} + k R^2 \theta \dot{\theta} = 0$$

$$\therefore \frac{3}{2} (m R^2) \dot{\theta} + k R^2 \theta = 0$$

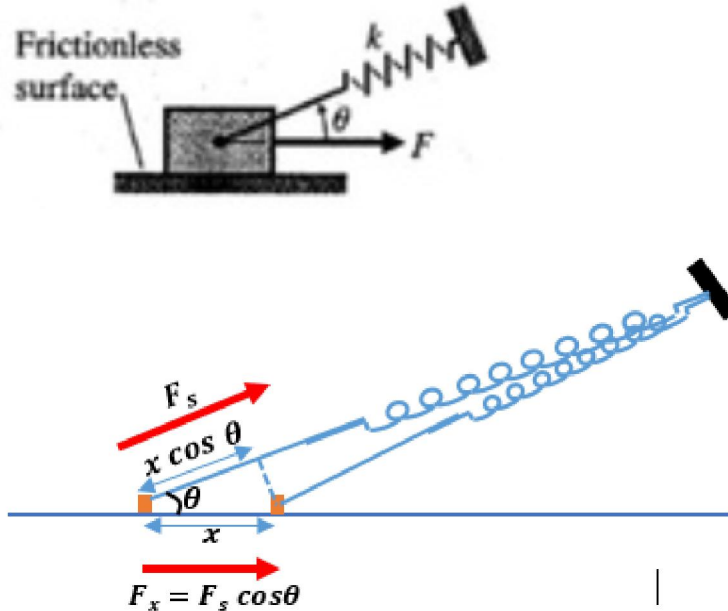
$$\text{The eq of motion, } \ddot{\theta} + \frac{k R^2}{\frac{3}{2} m R^2} \theta = 0$$

$$\omega_n = \sqrt{\left[\frac{k R^2}{\frac{3}{2} m R^2} \right]} \text{ rad/s}$$

$$f_n = \frac{1}{2\pi} \sqrt{\frac{2k}{3m}} \text{ Hz}$$

$$= \frac{1}{2\pi} \sqrt{\frac{2 \times 4000}{3 \times 4}} = 4.11 \text{ Hz}$$

EQUIVALENT STIFFNESS OF AN INCLINED SPRING



Let the block be given an small arbitrary displacement x along the horizontal surface. This causes the spring to elongate by a distance $x \cos \theta$. This elongation of the spring produces a restoring force F_s in the spring. Thus

$$F_s = -k x \cos \theta \quad (1)$$

The component F_x of spring force along the horizontal surface is $F_s \cos \theta$ (F_x is represented by F in your diagram).

$$F_x = F_s \cos \theta$$

Or

$$F_x = -k x \cos \theta \cos \theta = -k \cos^2 \theta x \quad (2)$$

In arriving at the above result we substituted F_s by $-k x \cos \theta$ from equation (1).

If the equivalent spring constant (stiffness) is k_{eq} then

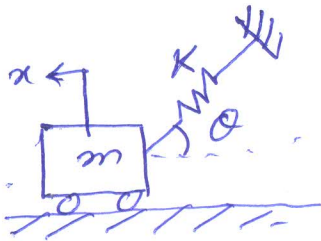
$$F_x = -k_{eq} x \quad (3)$$

Compare equations (2) and (3) to have

$$k_{eq} = k \cos^2 \theta$$

Vibration, Natural Frequency.

- Q. Determine the natural frequency of the spring-mass system shown below. Given $K = 1000 \text{ N/m}$
 $m = 10 \text{ kg}$, $\theta = 60^\circ$



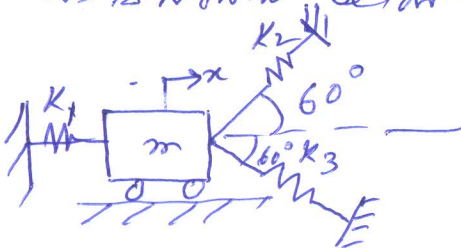
Soln

$$\text{Here } K_{\text{ed}} = K \cos^2 \theta = 1000 \times \cos^2 60^\circ = 250 \text{ N/m}$$

$$\therefore f_n = \frac{1}{2\pi} \sqrt{\frac{K_{\text{ed}}}{m}} = \frac{1}{2\pi} \sqrt{\frac{250}{10}} = 0.8 \text{ cycles/sec} \\ = 0.8 \text{ Hz Ans}$$

————— x —————

- Q. Determine the natural freq of the spring mass system as shown below. Given $K_1 = K_2 = K_3 = 1000 \text{ N/m}$, $m = 10 \text{ kg}$



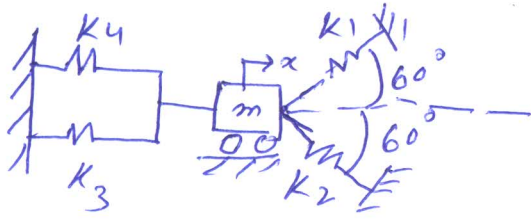
Soln All the springs are in parallel combination.

$$\text{Here } K_{\text{ed}} = K_1 + K_2 \cos^2 60^\circ + K_3 \cos^2 60^\circ \\ = 1000 + 2 \times 1000 \times \cos^2 60^\circ \\ = 1500 \text{ N/m}$$

$$\therefore f_n = \frac{1}{2\pi} \sqrt{\frac{K_{\text{ed}}}{m}} = \frac{1}{2\pi} \sqrt{\frac{1500}{10}} = 1.95 \text{ Hz Ans}$$

————— x —————

Q. Determine the natural freq of the spring mass system as shown below. Given $K_1 = K_4 = 100 \text{ N/m}$, $K_3 = K_4 = 50 \text{ N/m}$, $m = 10 \text{ kg}$



Soln The springs K_1, K_2, K_4, K_3 are in parallel.

$$K_{12} = K_1 \cos^2 60^\circ + K_2 \cos^2 60^\circ = 100 \cos^2 60^\circ + 100 \cos^2 60^\circ = 50 \text{ N/m}$$

$$K_{34} = K_3 + K_4 = 50 + 50 = 100 \text{ N/m}$$

Again K_{12} & K_{34} are in parallel combination

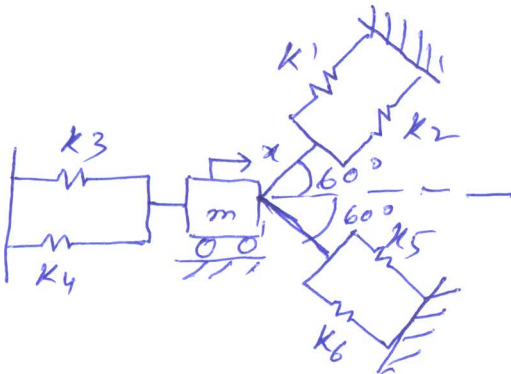


$$\therefore K_{eq} = K_{34} + K_{12} = 100 + 50 = 150 \text{ N/m}$$

$$\therefore f_n = \frac{1}{2\pi} \sqrt{\frac{K_{eq}}{m}} = \frac{1}{2\pi} \sqrt{\frac{150}{10}} = 0.62 \text{ cycles/sec} = \underline{0.62 \text{ Hz Ans.}}$$

— x —

Q. Determine the natural freq of the spring-mass system as shown below. Given $K_1 = K_2 = K_3 = K_4 = K_5 = K_6 = 100 \text{ N/m}$ mass $m = 10 \text{ kg}$.



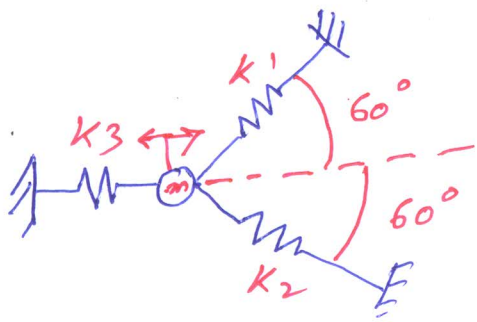
Note: - $K_{12} = K_1 + K_2$
 $K_{34} = K_3 + K_4$
 $K_{56} = K_5 + K_6$

Soln

$$K_{eq} = \text{~~150~~} 150 \text{ N/m}$$

$$f_n = \frac{1}{2\pi} \sqrt{\frac{K_{eq}}{m}} = \text{~~0.62~~} 0.62 \text{ Hz}$$

— x —



Direction of disturbance of m in x dir.

$$K_{ed} = k_1 \cos^2 60^\circ + k_2 \cos^2 60^\circ + k_3 \cos^2 0^\circ$$

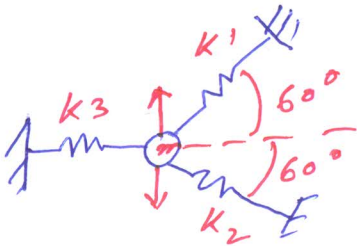
$$= k [0.25 + 0.25 + 1]$$

$$= k [1.50] = 100 \times 1.50 = 150 \text{ N/m}$$

$$f_n = \frac{1}{2\pi} \sqrt{\frac{K_{ed}}{m}} = \frac{1}{2\pi} \sqrt{\frac{150}{10}} = \underline{\underline{0.617 \text{ Hz}}}$$

Let $k_1 = k_2 = k_3 = 100 \text{ N/m}$

$m = 10 \text{ kg}$, $f_n = ?$



Direction of disturbance of m in y dir

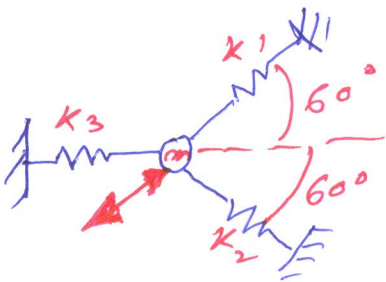
$$K_{ed} = k_3 \cos^2 90^\circ + k_2 \cos^2 30^\circ + k_1 \cos^2 30^\circ$$

$$= k [0 + 0.75 + 0.75] = k [1.50] \text{ N/m}$$

$$f_n = \frac{1}{2\pi} \sqrt{\frac{K_{ed}}{m}} = \underline{\underline{0.617 \text{ Hz}}}$$

Let $k_1 = k_2 = k_3 = 100 \text{ N/m}$

$m = 10 \text{ kg}$, $f_n = ?$



Direction of disturbance of m in direction of k_1 .

$$K_{ed} = k_1 \cos^2 0^\circ + k_2 \cos^2 60^\circ + k_3 \cos^2 60^\circ$$

$$= k [1 + 0.25 + 0.25] = k [1.50] \text{ N/m}$$

Let $k_1 = k_2 = k_3 = 100 \text{ N/m}$

$m = 10 \text{ kg}$, $f_n = ?$

$$f_n = \frac{1}{2\pi} \sqrt{\frac{K_{ed}}{m}} = \underline{\underline{0.617 \text{ Hz}}}$$

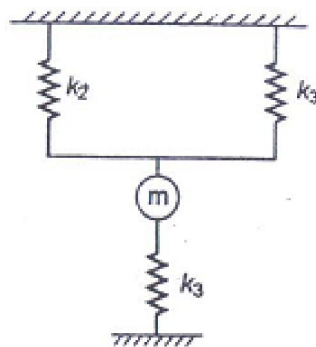
Conclusion: The natural frequency of the system is independent of direction of movement of mass m.

Model Questions on Natural Frequency of a Spring Mass System

A vibration system consists of a mass m and a spring of stiffness k . It has natural frequency of 12 Hz. The natural frequency decreases by 2 Hz. When an extra mass of 2 kg is attached to m . Find k and m .

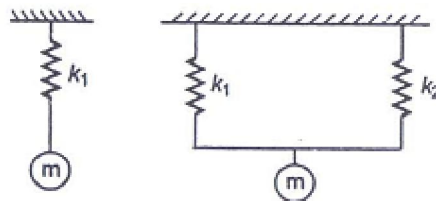
(Ans 25.783 N/mm)

Find the natural frequency of the system shown in Figure 1 if $k_1 = k_2 = 1500$ N/m, $k_3 = 2000$ N/m and $m = 5$ kg.



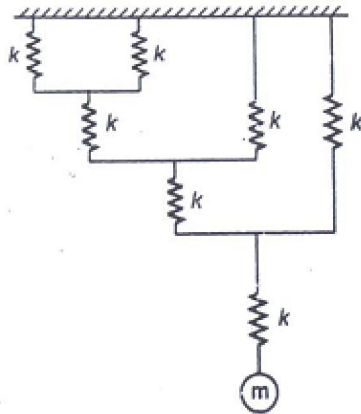
(Ans 5.03 Hz)

A Spring mass system (k_1 , m_1) has a natural frequency of f_1 as shown in figure 2. Another spring k_2 is connected in parallel to k_1 and the natural frequency increases by 30%. Find k_2 .



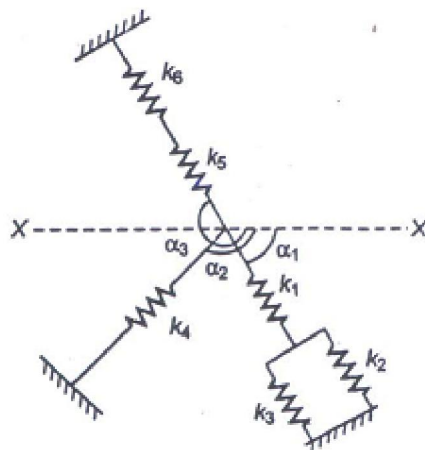
(Ans $k_2 = 0.69k_1$)

Calculate the natural frequency of the system shown if $k = 2 \times 10^5$ N/m and $m = 20$ kg.



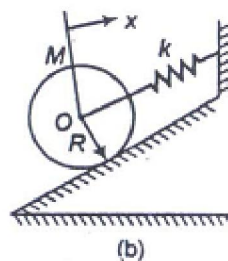
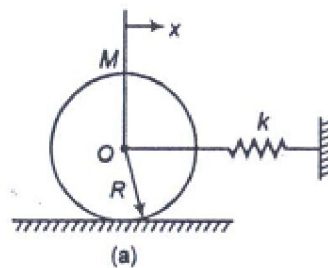
(Ans $f_n = 12.5 \text{ Hz}$)

Find the equivalent spring constant and natural frequency for oscillation of mass m in the direction $x-x$ as shown in figure.



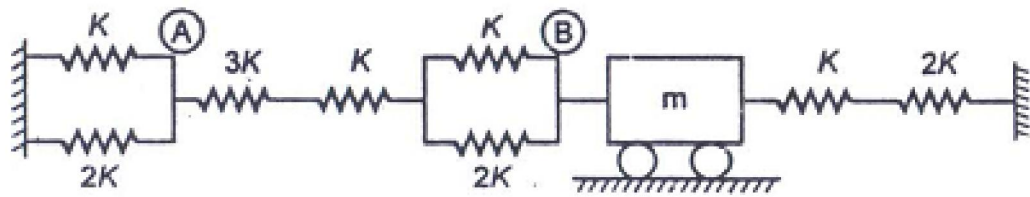
(Ans $f_n = \frac{1}{2\pi} \sqrt{\frac{k_e}{m}}$ $k_e = \frac{k_1 (k_2 + k_3)}{k_1 + k_2 + k_3} \cos^2 \alpha_1 + k_4 \cos^2 \alpha_2 + \frac{k_5 k_6}{k_5 + k_6} \cos^2 \alpha_3$)

A circular cylinder of mass ' m ' and radius ' R ' is connected by a spring of stiffness ' k ' as shown in Fig. (a). If it is free to roll on the rough surface which is horizontal without slipping. Find its natural frequency. What will happen to frequency if the system is placed on an inclined as shown in Fig. (b).



Ans. $f_n = \frac{1}{2\pi} \sqrt{\frac{2k}{3M}} \text{ CPS.}$

Determine the equivalent stiffness of the system as shown in Fig.



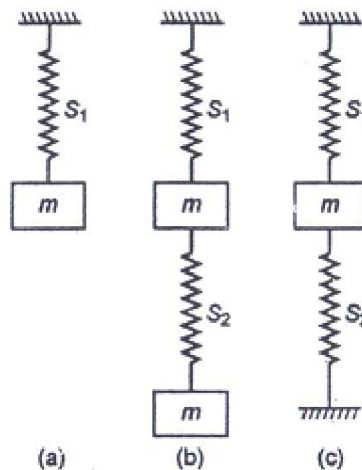
$$k_{eq} = 1.18 \text{ N/m}$$

A

Determine the equivalent spring stiffness and the natural frequency of the following vibrating systems when

- (a) mass is suspended to a spring
- (b) mass is suspended at the bottom of two springs in series
- (c) mass is fixed in between two springs

Take $S_1 = 5 \text{ N/mm}$, $S_2 = 8 \text{ N/mm}$, $m = 10 \text{ kg}$,



- Ans a) $f_n = 3.56 \text{ Hz}$
 b) $f_n = 2.79 \text{ Hz}$
 c) $f_n = 5.74 \text{ Hz}$